Student Academic Life style Research

Project Overview

- *This project analyzes student lifestyle factors and learning styles to understand their impact on academic performance.*
- *Using Excel and Power BI, the study applies data cleaning and comparative evaluation techniques to derive meaningful insights.*

Objective

- *Clean and structure the dataset.*
- *Analyze student lifestyle patterns.*
- *Compare academic performance across learning styles and gender.*
- *Identify factors influencing exam scores.*
- *Provide data-driven recommendations.*

Data Sources

- ***Source Description:*** *The dataset was extracted from DeepDataLake.com for the purpose of academic analysis.*
- ***Timeline:*** *2023-2024*
- ***Domain:*** *Education*

Problem Statement

- *To analyze students' lifestyles in relation to their academic performance and daily habits.*
- *To analyze students who appeared for examinations based on age groups.*
- *To evaluate students' academic performance.*
- *To examine students' lifestyle factors.*
- *To identify highly certified learners.*
- *To determine students with 100% attendance.*
- *To determine the number of top-performing students.*
- *To identify which lifestyle patterns are unsuitable for students.*
- *To determine which habits positively support academic performance.*
- *To examine the impact of stress on students' academic achievement.*

Attribute (Column /Features) Details:

Attribute Name	*Data Type*	*Description*
Student ID	*General*	*Unique identifier for each Student*
Age	*Number*	*Age of the Student*
Gender	*Text*	*Gender Identity*
Study Hours per Week	*Number*	*Total Hours studied per week by each student*
Preferred Learning Style	*Text*	*Preferred Learning Style of each Student*
Online Courses Completed	*Number*	*Total number of online course completed by each student*
Participation in Discussions	*Text*	*Whether the student participated in Discussion or not - Yes/No*
Assignment Completion Rate (%)	*Number*	*Overall assignment completion rate of each student*
Exam Score (%)	*Number*	*Mark scored by each student*
Attendance Rate (%)	*Number*	*Overall attendance rate of each student*
Use of Educational Tech	*Text*	*Whether the student used Educational Tech or not – Yes/No*
Self Reported Stress Level	*Text*	*Stress level of each student – High, Medium, Low*
Time Spent on Social Media (hours/week)	*Number*	*Total time spent on social media by each student*
Sleep Hours per Night	*Number*	*Total sleeping hours per night of each student*
Final Grade	*Text*	*Grade based on the marks scored in the Exam – A,B,C,D*

Tools & Technologies

Excel: *Data cleaning, transformation, Sorting, Created separate tables using filters, Pivot Tables and Chart*

Power BI: *Data modeling, DAX measures & Visualization*

Data Pre-Processing (Excel)

Tasks Performed:

Data Cleaning & Transformation:

- *The data type of each column has been updated according to the specifications in the attribute section.*
- *The categorical values in the columns Participation in discussions and Use of educational Tech were converted into numerical form (Yes = 1, No = 0).*

Filtering & Sorting:

- *The dataset was arranged in ascending order based on Student ID for better organization.*
- *Filtering techniques were applied to detect missing values, duplicate entries, and data inconsistencies.*
- *The dataset was segmented using filtering techniques to create distinct tables for each learning style (Kinesthetic, Reading/Writing, Auditory and Visual).*

Pivot Tables:

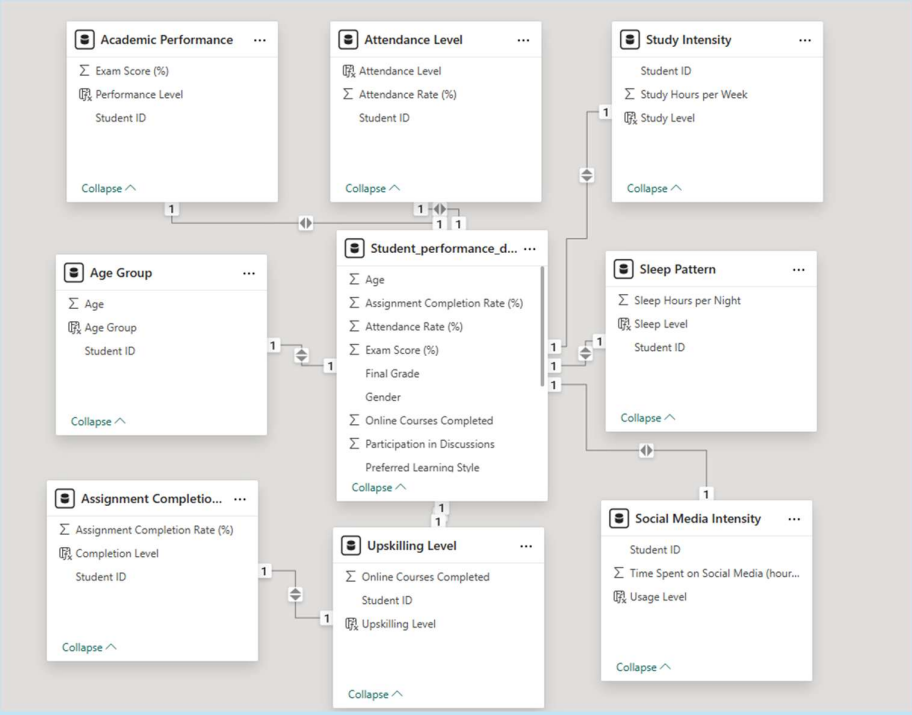
- *Four pivot tables were generated by segmenting the dataset according to learning style categories.*
- *The relationship between stress level and learning style was evaluated through pivot table analysis.*
- *The pivot table incorporated key performance indicators to generate preliminary insights, including number of students, total online courses completed, and averages of social media usage, study hours, sleep duration, attendance rate, assignment completion rate, and exam score.*
- *A comparative analysis was conducted across the four learning styles based on student count and average values of study hours, social media usage, sleep duration, and exam scores.*
- *A gender-based analysis was conducted to examine the relationship between online course completion and academic grade.*

Chart:

- *A comparative chart was created to display the total number of online courses completed by each gender within different grade levels.*

Data Modelling and DAX (Power BI)

Data Model: *Designed and developed 8 relational tables and established necessary relationships to ensure data integrity.*



Calculated Columns & DAX Measures:

Caculated table: DAX formula

- ***Sleep Pattern*** =

```
SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Sleep Hours per Night]
```

Student ID	Sleep Hours per Night
S00001	8
S00002	8
S00003	7
S00004	10
S00005	8
S00006	8
S00007	10

- ***Social Media Intensity*** =
  SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Time Spent on Social Media (hours/week)])

Student ID	Time Spent on Social Media (hours/week)
S00001	9
S00002	28
S00003	13
S00004	24
S00005	26
S00006	25
S00007	30
S00008	4
S00009	26

- ***Study Intensity*** =
  SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Study Hours per Week])

Student ID	Study Hours per Week
S00001	48
S00002	30
S00003	47
S00004	13
S00005	24
S00006	26
S00007	49
S00008	14
S00009	45
S00010	35

- ***Upskilling Level*** =
  SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Online Courses Completed])

Student ID	Online Courses Completed
S00001	14
S00002	20
S00003	11
S00004	0
S00005	19
S00006	5
S00007	13
S00008	5
S00009	16

- ***Academic Performance*** =
```
SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Participation in Discussions],Student_performance_dataset[Use of Educational Tech],Student_performance_dataset[Student ID],Student_performance_dataset[Exam Score (%)])
```

Student ID	Exam Score (%)	Participation in Discussions	Use of Educational Tech
S00001	69	1	1
S00002	40	0	1
S00003	43	0	1
S00004	70	1	1
S00005	63	1	1
S00006	54	1	0
S00007	44	1	1
S00008	56	1	1

- ***Age Group*** =
```
SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Age])
```

Student ID	Age
S00001	18
S00002	29
S00003	20
S00004	23
S00005	19
S00006	28
S00007	19
S00008	27
S00009	22

- ***Assignment Completion Level*** =
```
SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Assignment Completion Rate (%)])
```

Student ID	Assignment Completion Rate (%)
S00001	100
S00002	71
S00003	60
S00004	63
S00005	59
S00006	63
S00007	91
S00008	88
S00009	52

- ***Attendance Level*** =
```
SUMMARIZE(Student_performance_dataset,Student_performance_dataset[Student ID],Student_performance_dataset[Attendance Rate (%)])
```

Student ID	Attendance Rate (%)
S00001	66
S00002	57
S00003	79
S00004	60
S00005	93
S00006	80
S00007	66

Caculated Column: DAX formula

- ***Sleep Level*** =
```
SWITCH(TRUE(),
'Sleep Pattern'[Sleep Hours per Night]<6,"Insufficient",
'Sleep Pattern'[Sleep Hours per Night]>=6 && 'Sleep Pattern'[Sleep Hours per Night]<=8,"Adequate",
"Extended")
```

Student ID	Sleep Hours per Night	Sleep Level
S00001	8	Adequate
S00002	8	Adequate
S00003	7	Adequate
S00004	10	Extended
S00005	8	Adequate
S00006	8	Adequate
S00007	10	Extended

- ***Usage Level*** =
```
SWITCH(TRUE(),
'Social Media Intensity'[Time Spent on Social Media (hours/week)]<3,"Minimal User",
'Social Media Intensity'[Time Spent on Social Media (hours/week)]>3 && 'Social Media Intensity'[Time Spent on Social Media (hours/week)]<=10,"Casual User",
'Social Media Intensity'[Time Spent on Social Media (hours/week)]>10 && 'Social Media Intensity'[Time Spent on Social Media (hours/week)]<=20,"Regular User",
"Excessive User")
```

Student ID	Time Spent on Social Media (hours/week)	Usage Level
S00001	9	Casual User
S00002	28	Excessive User
S00003	13	Regular User
S00004	24	Excessive User
S00005	26	Excessive User
S00006	25	Excessive User
S00007	30	Excessive User
S00008	4	Casual User
S00009	26	Excessive User

- ***Study Level*** = SWITCH(TRUE(),
'Study Intensity'[Study Hours per Week]<=10,"Minimal",
'Study Intensity'[Study Hours per Week]>10 && 'Study Intensity'[Study Hours per Week]<=20,"Light",
'Study Intensity'[Study Hours per Week]>20 && 'Study Intensity'[Study Hours per Week]<=30, "Regular",
'Study Intensity'[Study Hours per Week]>30 && 'Study Intensity'[Study Hours per Week]<=40, "Dedicated",
"Intensive")

Student ID	Study Hours per Week	Study Level
S00001	48	Intensive
S00002	30	Regular
S00003	47	Intensive
S00004	13	Light
S00005	24	Regular
S00006	26	Regular
S00007	49	Intensive
S00008	14	Light
S00009	45	Intensive
S00010	35	Dedicated

- ***Upskilling Level*** = SWITCH(TRUE(),
'Upskilling Level'[Online Courses Completed]<=2,"Beginner",
'Upskilling Level'[Online Courses Completed]>2 && 'Upskilling Level'[Online Courses Completed]<=10,"Developing",
'Upskilling Level'[Online Courses Completed]>10 && 'Upskilling Level'[Online Courses Completed]<=15,"Advanced",
"Expert")

Student ID	Online Courses Completed	Upskilling Level
S00001	14	Advanced
S00002	20	Expert
S00003	11	Advanced
S00004	0	Beginner
S00005	19	Expert
S00006	5	Developing
S00007	13	Advanced
S00008	5	Developing
S00009	16	Expert

- ***Performance Level*** = SWITCH(TRUE(),
'Academic Performance'[Exam Score (%)] >=80,"Top Performer",
'Academic Performance'[Exam Score (%)]>=60 && 'Academic Performance'[Exam Score (%)]<80, "Average Performer",
"Low Performer")

Student ID	Exam Score (%)	Performance Level
S00001	69	Average Performer
S00002	40	Low Performer
S00003	43	Low Performer
S00004	70	Average Performer
S00005	63	Average Performer
S00006	54	Low Performer
S00007	44	Low Performer
S00008	56	Low Performer
S00009	78	Average Performer
S00010	55	Low Performer
S00011	44	Low Performer
S00012	56	Low Performer
S00013	73	Average Performer
S00014	60	Average Performer
S00015	71	Average Performer
S00016	90	Top Perfomer

- ***Academic Interaction*** = IF('Academic Performance'[Participation in Discussions]=1 && 'Academic Performance'[Use of Educational Tech]=1,"Active","Passive")

Participation in Discussions	Use of Educational Tech	Academic Interaction
1	1	Active
0	1	Passive
0	1	Passive
1	1	Active
1	1	Active
1	0	Passive
1	1	Active
1	1	Active

- ***Age Group*** = SWITCH(TRUE(),
'Age Group'[Age]>=18 && 'Age Group'[Age]<=20,"Sub-Junior",
'Age Group'[Age]>20 && 'Age Group'[Age]<=25,"Junior",
"Senior")

Student ID	Age	Age Group
S00001	18	Sub-Junior
S00002	29	Senior
S00003	20	Sub-Junior
S00004	23	Junior
S00005	19	Sub-Junior
S00006	28	Senior
S00007	19	Sub-Junior
S00008	27	Senior
S00009	22	Junior

- ***Completion Level*** = SWITCH(TRUE(),
'Assignment Completion Level'[Assignment Completion Rate (%)]<=60,"Very Low",
'Assignment Completion Level'[Assignment Completion Rate (%)]>60 && 'Assignment Completion Level'[Assignment Completion Rate (%)]<=70,"Low",
'Assignment Completion Level'[Assignment Completion Rate (%)]>70 && 'Assignment Completion Level'[Assignment Completion Rate (%)]<=80,"Moderate",
'Assignment Completion Level'[Assignment Completion Rate (%)]>80 && 'Assignment Completion Level'[Assignment Completion Rate (%)]<=90,"High",
"Excellent")

Student ID	Assignment Completion Rate (%)	Completion Level
S00001	100	Excellent
S00002	71	Moderate
S00003	60	Very Low
S00004	63	Low
S00005	59	Very Low
S00006	63	Low
S00007	91	Excellent
S00008	88	High
S00009	52	Very Low

- ***Attendance Level*** = SWITCH(TRUE(),
'Attendance Level'[Attendance Rate (%)]<=70,"Very Low",
'Attendance Level'[Attendance Rate (%)]>70 && 'Attendance Level'[Attendance Rate (%)]<=80,"Low",
'Attendance Level'[Attendance Rate (%)]>80 && 'Attendance Level'[Attendance Rate (%)]<=90,"Moderate",
'Attendance Level'[Attendance Rate (%)]>90 && 'Attendance Level'[Attendance Rate (%)]<=99,"High",
"Excellent")

Student ID	Attendance Rate (%)	Attendance Level
S00001	66	Very Low
S00002	57	Very Low
S00003	79	Low
S00004	60	Very Low
S00005	93	High
S00006	80	Low
S00007	66	Very Low
S00008	76	Low
S00009	70	Very Low
S00010	100	Excellent

Calculated Measures: DAX formula

- ***Total Students*** = COUNT(Student_performance_dataset[Student ID])

Total Students

10K

- ***Top Performers*** = CALCULATE(COUNTROWS('Academic Performance'),filter('Academic Performance','Academic Performance'[Performance Level]="Top Performer"))

Top Performers

3475

- ***Minimal SM Users*** = CALCULATE(COUNTROWS('Social Media Intensity'),FILTER('Social Media Intensity','Social Media Intensity'[Usage Level]="Minimal User"))

Minimal SM Users

1004

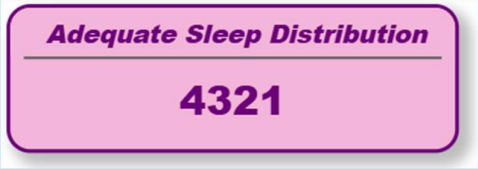
- ***Intensive Study Group*** = CALCULATE(COUNTROWS('Study Intensity'),filter('Study Intensity','Study Intensity'[Study Level]="Intensive"))

Intensive Study Group

2037

- ***Highly Certified Learners*** = CALCULATE(COUNTROWS('Upskilling Level'),filter('Upskilling Level','Upskilling Level'[Upskilling Level]="Expert"))

- ***Adequate Sleep Distribution*** = CALCULATE(COUNTROWS('Sleep Pattern'),FILTER('Sleep Pattern','Sleep Pattern'[Sleep Level]="Adequate"))



- ***100% Attendance*** = CALCULATE(COUNTROWS('Attendance Level'),FILTER('Attendance Level','Attendance Level'[Attendance Level]="Excellent"))

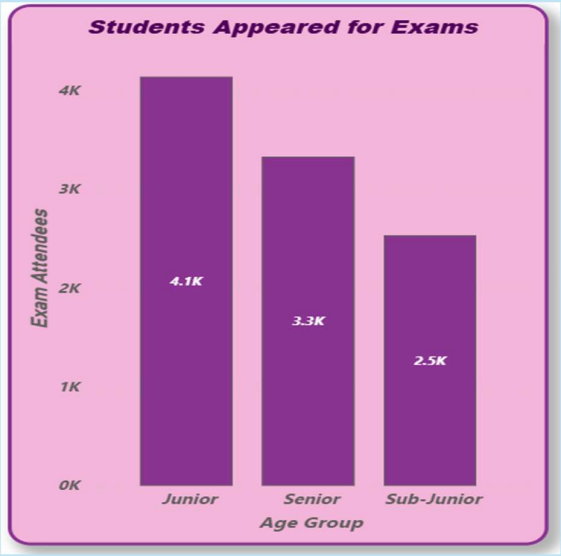
Analysis and Visualizations (Power BI)

Analysis:

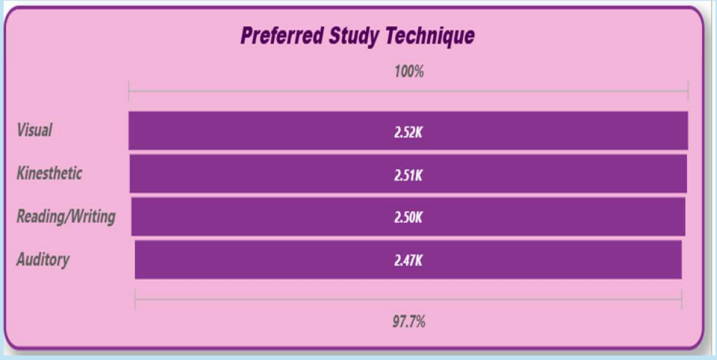
- *Student Academic Record*
- *Student Activity Record*
- *Comparative Analysis of Students' Life Style*
- *Age-Wise Stress Analysis*

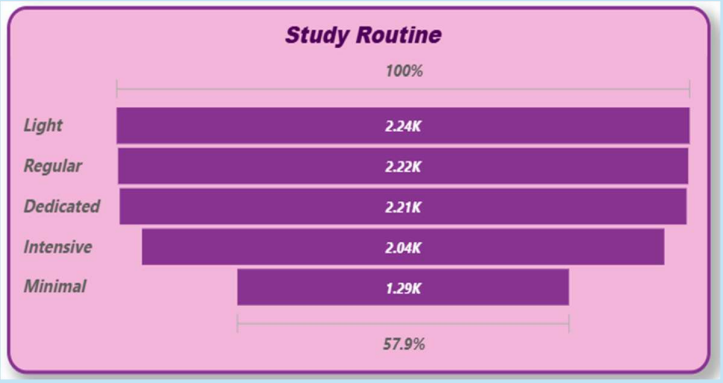
Mutliple Visualizations:

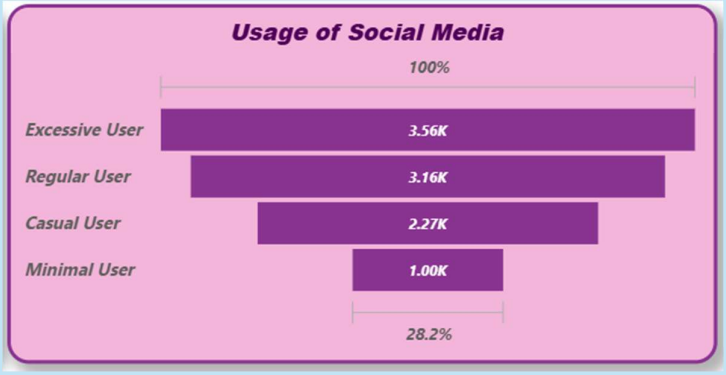
- ***Column Chart:***

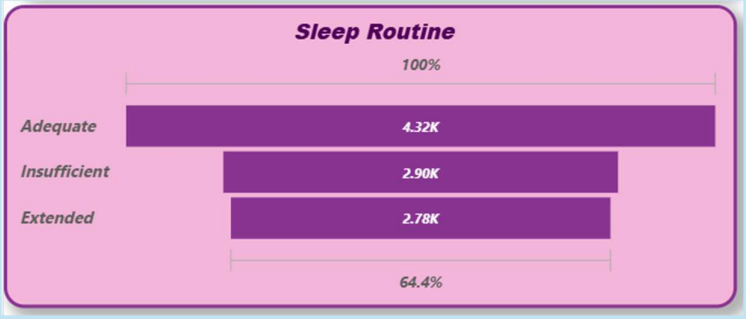


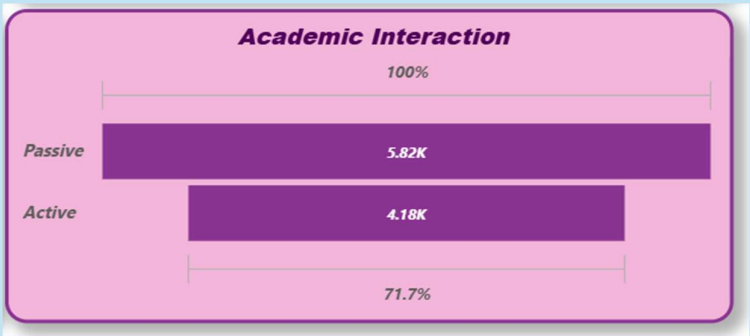
- ***Funnel Chart:***



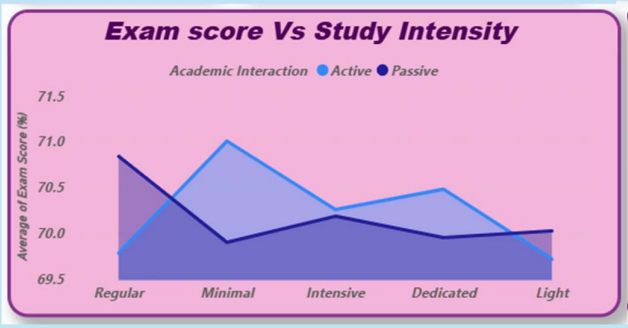


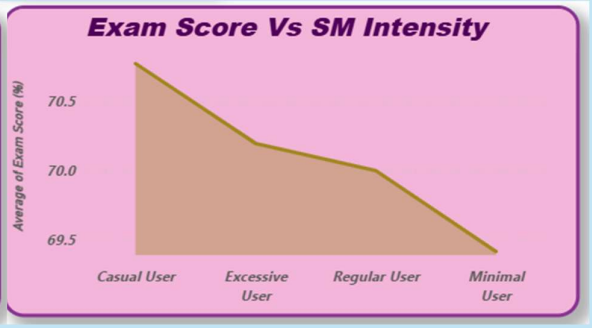


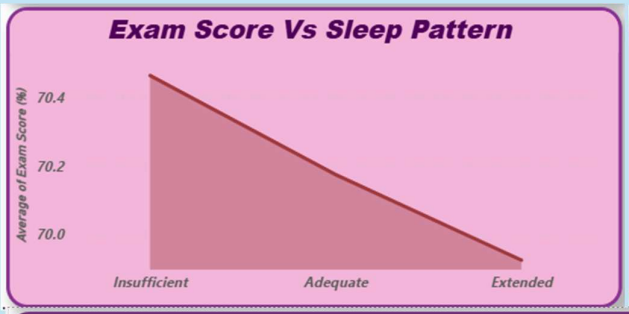


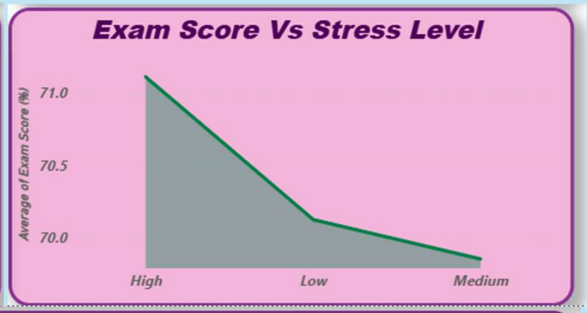


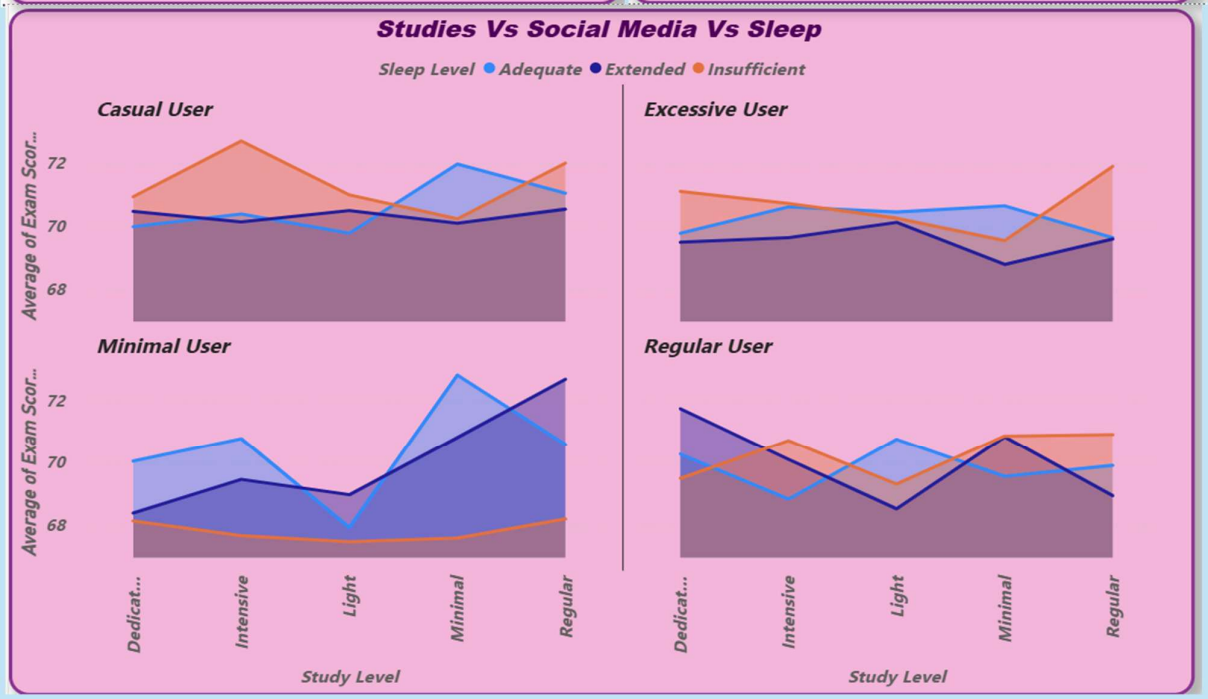
- ***Area Chart:***



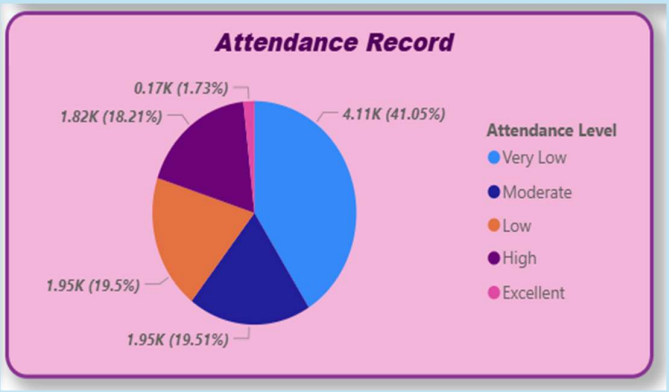






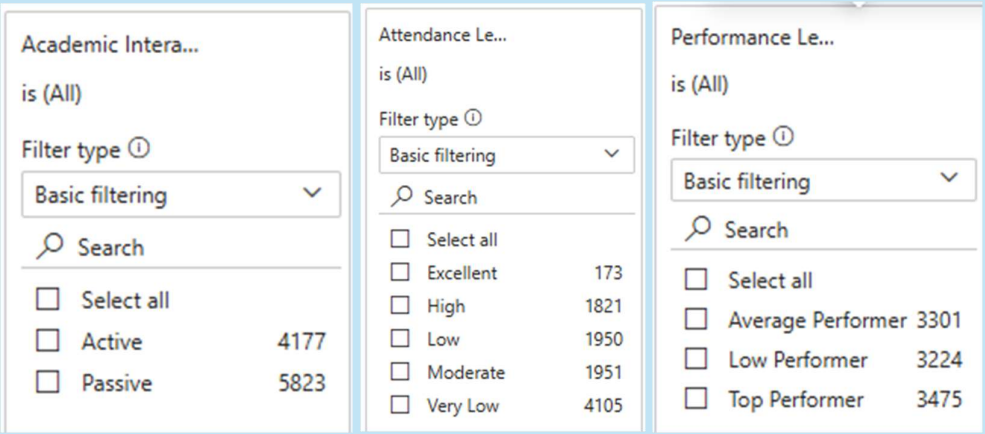


- ***Pie-Chart:***



- ***Slicers:***

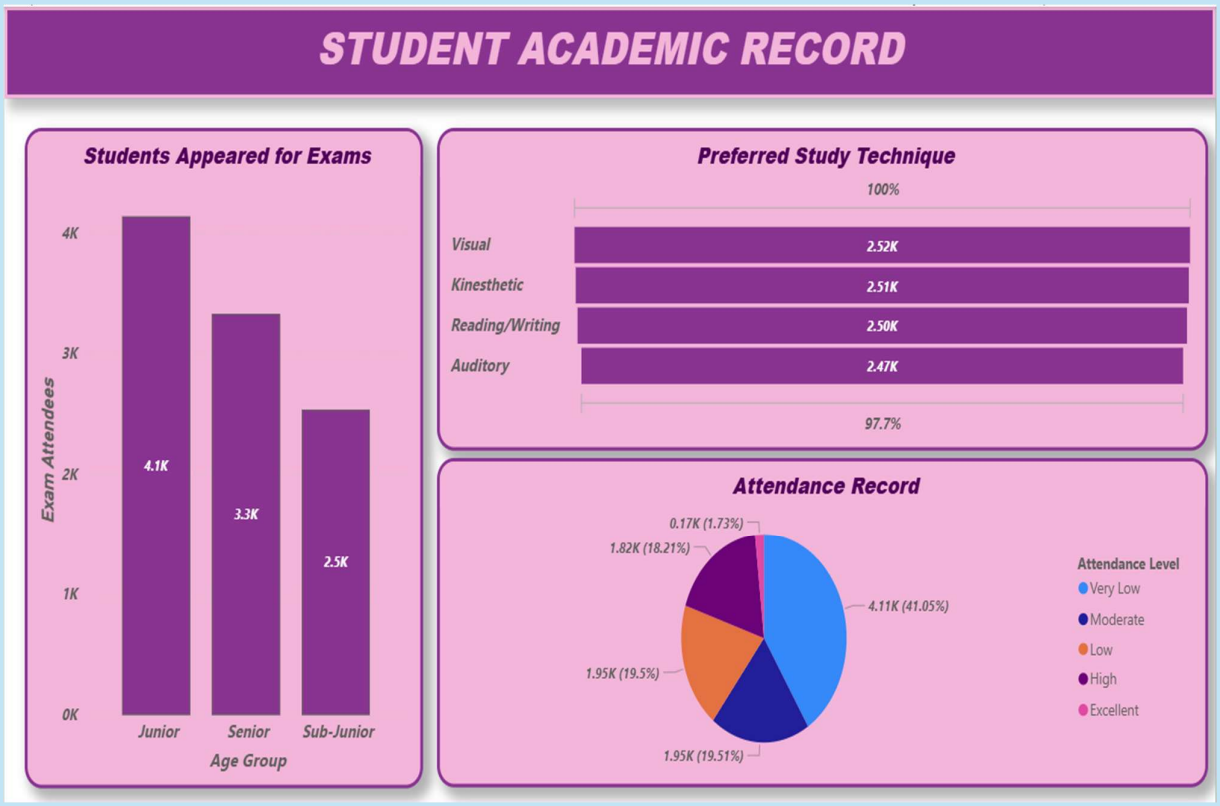
- ***Filters:***

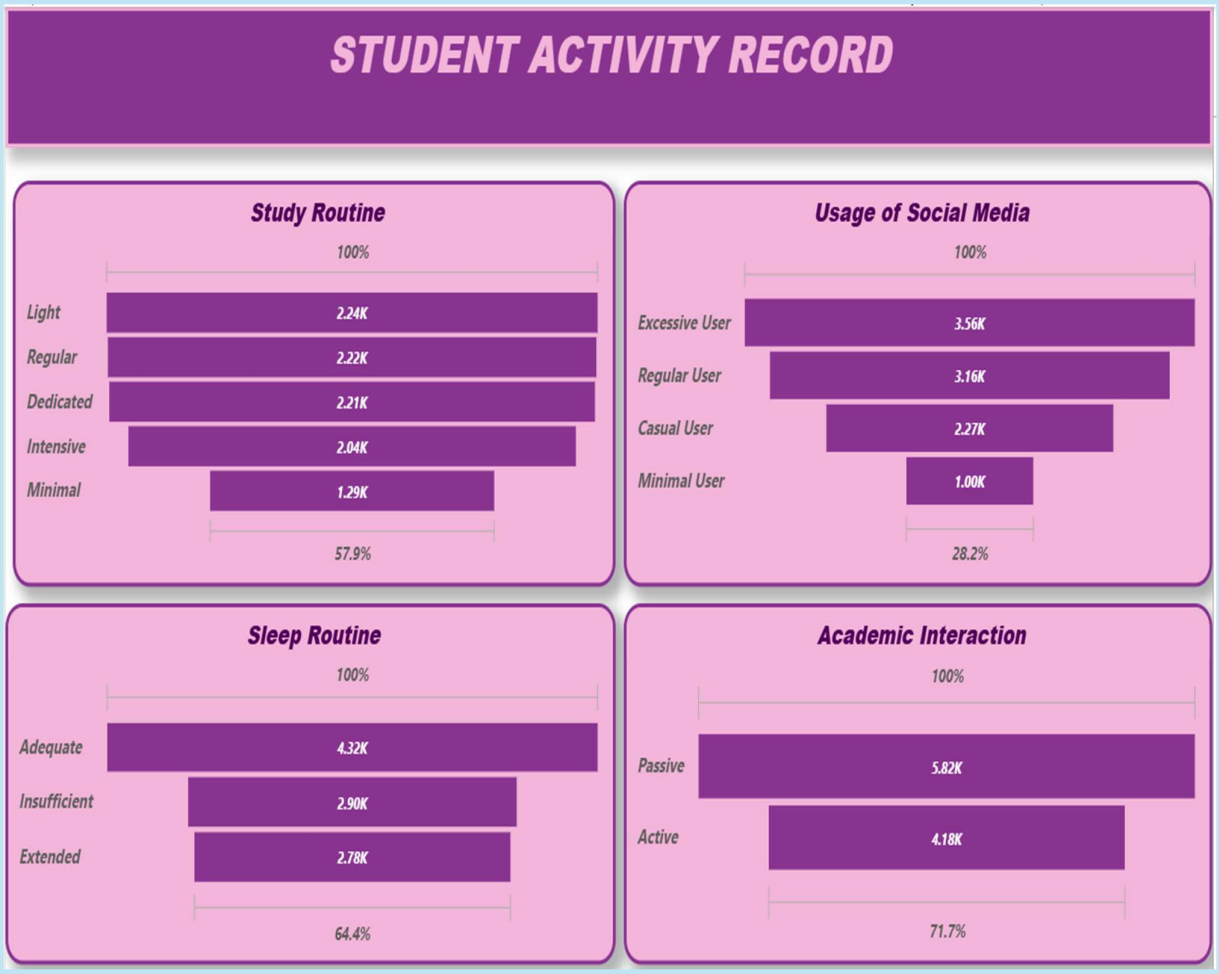


- ***Cards:***

- ***Book mark/Buttons:***

Dashboards:


STUDENT ACADEMIC RECORD
Students Appeared for Exams
Exam Attendees
4K
3K
2K
1K
0K
4.1K
3.3K
2.5K
Junior
Senior
Sub-Junior
Age Group
Preferred Study Technique
100%
Visual
2.52K
Kinesthetic
2.51K
Reading/Writing
2.50K
Auditory
2.47K
97.7%
Attendance Record
0.17K (1.73%)
1.82K (18.21%)
4.11K (41.05%)
1.95K (19.5%)
1.95K (19.51%)
Attendance Level
Very Low
Moderate
Low
High
Excellent

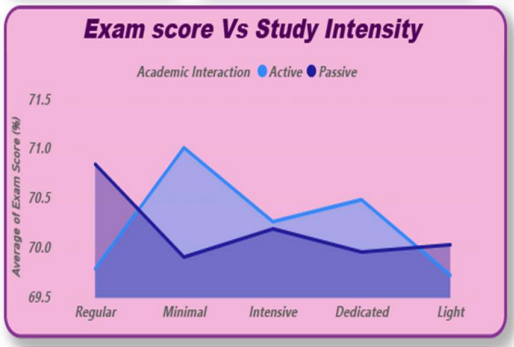

STUDENT ACTIVITY RECORD
Study Routine
100%
Light
2.24K
Regular
2.22K
Dedicated
2.21K
Intensive
2.04K
Minimal
1.29K
57.9%
Usage of Social Media
100%
Excessive User
3.56K
Regular User
3.16K
Casual User
2.27K
Minimal User
1.00K
28.2%
Sleep Routine
100%
Adequate
4.32K
Insufficient
2.90K
Extended
2.78K
64.4%
Academic Interaction
100%
Passive
5.82K
Active
4.18K
71.7%

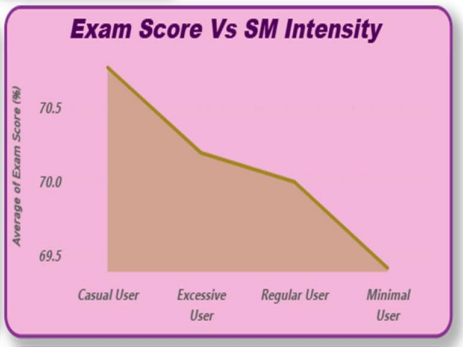
Comparative Analysis of Student's Life Style

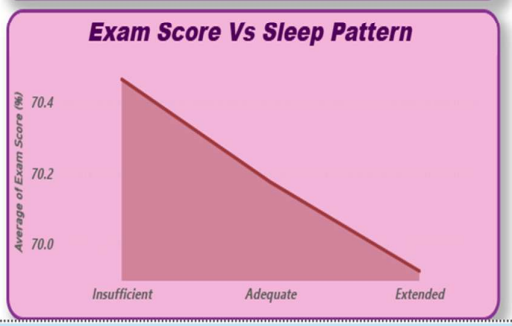
Total Students	Highly Certified Learners	100% Attendance	Top Performers
10K	2457	173	3475

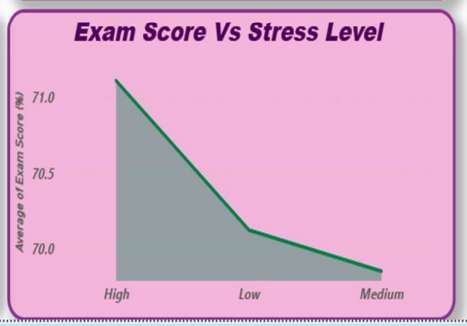
Overall Best Performers

Low Performers

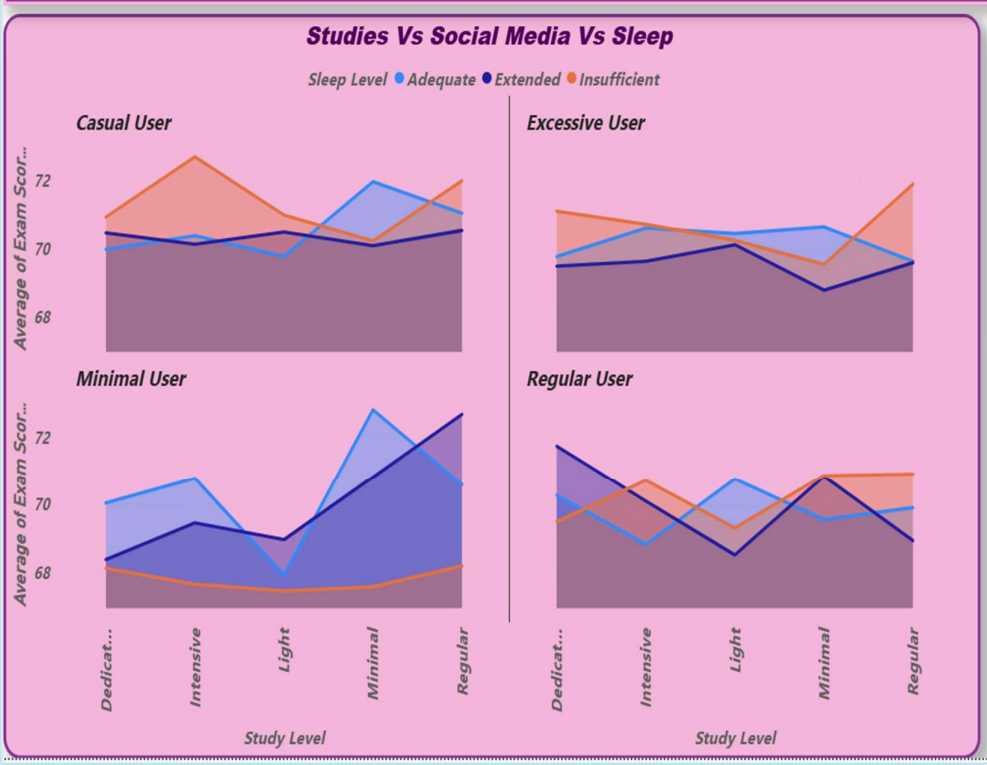


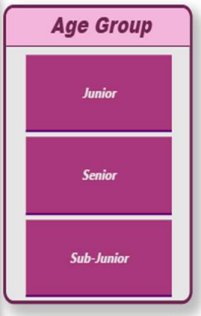


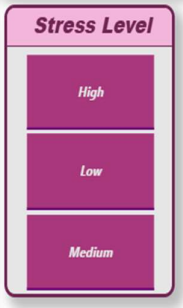




Age-Wise Stress Analysis







Insights

Student Academic Record

- *The total number of students is 10,000, with the majority aged between 21 and 25.*
- *Students prefer various study techniques: Visual (2,520), Kinesthetic (2,510), Reading/Writing (2,500), and Auditory (2,470), with most students favoring the Visual method.*

Student Activity Record

- *Most students follow a study routine of 11–20 hours per week and spend 11–20 hours per week on screens.*
- *The majority maintain adequate sleep, and 5,820 students actively participate in classroom discussions and use educational tools.*

Comparative Analysis of Student Lifestyle

- *Among the students, there are 3,475 top performers, 173 students with 100% attendance, and 2,457 highly certified learners.*
- *The intensive study group studies 31–40 hours per week, while 4,321 students get adequate sleep (6–8 hours) and 1,004 students maintain minimal screen time (<3 hours per week).*
- *The best student habits include continuous learning, regular class attendance, active participation, use of educational tools, dedication to studies, limited screen time, completing assignments properly, and using hands-on/kinesthetic learning methods.*
- *Conversely, the worst habits include completing fewer online courses, being inactive in class, not using educational tools, sleeping excessively or insufficiently, irregular study habits, poor assignment completion, low attendance, and relying only on the Visual study method.*

Age-wise Stress Analysis

- *Stress levels are higher among sub-junior students compared to other age groups.*
- *While stress does not significantly impact academic performance, it is often linked to insufficient or excessive sleep.*

Conclusions

- *Encourage structured study routines to balance workload and improve performance.*
- *Promote healthy sleep habits (6–8 hours) to reduce stress and support academics.*
- *Minimize non-study screen time (<3 hours/week) to improve focus.*
- *Increase student engagement in class discussions and use of educational tools.*
- *Encourage the use of multiple learning techniques beyond just Visual.*
- *Recognize and motivate top performers and highly certified learners.*
- *Promote positive habits such as regular attendance, continuous learning, hands-on study, and timely assignment completion.*

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